Reproduction of Merchant, Arnold, Baumgart, Mukherjee, Pein & Pinker (ISPSD 1991)

"Realization of High Breakdown Voltage (>700 V) in Thin SOI Devices"
Proc. ISPSD 1991, pp. 31–35

Independent derivation, simulation and figure-by-figure comparison,
with an extension to 4H-SiC and GaN films

Abstract

A lateral P^+–N SOI diode's breakdown voltage is maximized, for a given lateral depletion width W, by choosing the doping profile $N(x)$ that makes the lateral field E_x laterally uniform rather than the usual way around (doping → field). The paper derives the resulting closed-form field distribution, an avalanche-breakdown model via two dominant ionization paths, and shows — experimentally and theoretically — that breakdown voltage *increases* as the SOI film thins below ∼1 µm, because the silicon's own critical field rises faster than the shrinking ionization path length loses. This note reproduces the closed-form model (Figs. 3–5), and extends it, unchanged, to 4H-SiC and GaN films using literature Chynoweth ionization fits.

Key results (model vs. paper)

Quantity	Model	Paper
Oxide breakdown voltage, $t_s = 0.1\,\mu m$, $t_{ox} = 2\,\mu m$	1210 V	1200 V (Fig. 4 dashed line)
BV, $t_s = 0.1\,\mu m$, $t_{ox} = 2\,\mu m$, $W \to \infty$	863 V	∼865 V (Fig. 5 plateau)
BV, $t_s = 0.1\,\mu m$, $t_{ox} = 2\,\mu m$, $W = 70\,\mu m$	849 V	750 V (Fig. 3, $L = 70\,\mu m$ device)
Design W for BV = 750 V (Fig. 3)	49.4 µm	$< L = 70\,\mu m$ ✓

1 Theory

1.1 The device and the design goal

A lateral P^+–N diode sits in a thin SOI film over a buried oxide (BOX) on a grounded N^+ substrate, shorted to the P^+ region (Fig. 2). x is lateral (0 at the P^+/N junction), y vertical, downward. t_s, t_{ox} are the SOI and buried-oxide thickness, L the P^+-to-N^+ spacing, W the lateral depletion width at breakdown ($W < L$).

Unlike a conventional RESURF design, the paper works *backward*: it picks the field distribution that maximizes breakdown voltage for a given W — a laterally *uniform* E_x, independent of y — then derives the lateral doping $N(x)$ that produces it via Poisson's equation.

1.2 Closed-form field and potential

Gauss's law and the uniform-E_x condition (paper eqs. 2–4) give, with $E_y(x,0) = 0$ (full depletion, minimum surface field) and $\phi(x,0) = Vx/W$ (linear top-surface potential):

$$E_y(x,y) = \frac{V x y}{t\, W\, t_s}, \tag{1}$$

$$E_x(y) = -\frac{V}{W}\left(1 - \frac{y^2}{2 t_s t}\right), \tag{2}$$

$$t = \frac{t_s}{2} + \frac{K_s}{K_{ox}} t_{ox}, \qquad \frac{K_s}{K_{ox}} = \frac{11.7}{3.9} = 3, \tag{3}$$

for $0 \le x \le W$, $0 \le y \le t_s$. t is the effective series thickness of half-depleted silicon plus 3× the oxide (the Si/oxide displacement-continuity jump). E_x peaks at $y = 0$; E_y peaks at the corner

$(x = W, y = t_s)$. Integrating the resulting $Q(x)$ shows the required lateral doping is *linear* in x — the SOI analogue of the Stengl–Gösele "variation of lateral doping" concept the paper cites. The potential itself (used to reproduce Fig. 3) is

$$\phi(x, y) = \frac{Vx}{W}\left(1 - \frac{y^2}{2tt_s}\right), \qquad 0 \le y \le t_s, \tag{4}$$

continuing linearly in y through the charge-free oxide beneath.

1.3 Avalanche breakdown

Ionization occurs only in the silicon, via the paper's own Chynoweth fit (eq. 10):

$$\alpha(E) = a\, e^{-b/E}, \qquad a = 7.03 \times 10^5\,\mathrm{cm}^{-1},\ \ b = 1.468 \times 10^6\,\mathrm{V\,cm}^{-1} \tag{5}$$

(valid $1.75 \times 10^5 \le E \le 6.4 \times 10^5\,\mathrm{V\,cm}^{-1}$; a different b than Nakagawa 1990's high-field branch). Two generation points dominate:

Path 1 — at $y = 0$ where $|E_x|$ is largest (and, by design, constant across $0 < x < W$): identical to a 1-D p-i-n diode of width W,

$$I_1 = aW\, e^{-bW/V}. \tag{6}$$

Path 2 — at the corner $(x = W, y = t_s)$ where $|E_y|$ is largest: a vertical leg through the silicon depth (closed form via the exponential integral Ei) plus a horizontal leg back along $y = t_s$,

$$I_2' = at_s\gamma\Big[\mathrm{Ei}(-\gamma) + \frac{e^{-\gamma}}{\gamma}\Big], \qquad \gamma = \frac{bt}{V}, \tag{7}$$

$$I_2'' = aW \exp\!\left(-\frac{bW}{V(1 - t_s/2t)}\right). \tag{8}$$

Breakdown is the lowest V solving $(1-I_1)(1-I_2'-I_2'') = 0$, i.e. the lowest V at which $\max(I_1, I_2'+I_2'')$ first reaches 1.

1.4 Why thinner SOI can raise the breakdown voltage

As $W \to \infty$, I_1 and I_2'' vanish (they decay like $We^{-\mathrm{const}\cdot W}$), leaving only $I_2'(V, t_s, t_{ox}) = 1$ to solve — a $\mathrm{BV}(t_s, t_{ox})$ relation independent of W (Fig. 4). A shorter silicon path (smaller t_s) needs a *higher* critical field to reach ionization integral 1, exactly as in a one-sided step junction. Below $t_s \sim 1\,\mu\mathrm{m}$ this effect overtakes the "shorter path" effect, so BV rises again as t_s falls — the characteristic "U" shape of Fig. 4 — until the buried oxide's own dielectric strength ($\sim 600\,\mathrm{V\,\mu m}^{-1}$) caps the achievable BV.

2 Figure-by-figure comparison

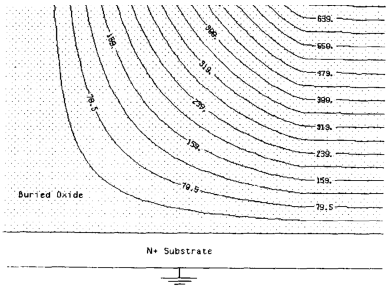
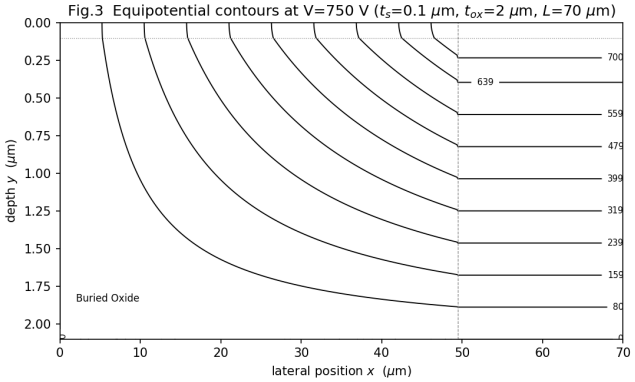


FIG. 3: Two-dimensional off-state numerical simulation of the SOI diode of Fig. 2 with t_s = 0.1 μm, t_{ox} = 2 μm, and L = 70 μm (plot not to scale). Equipotential contours are shown for reverse bias voltage of 750 V, which, as determined by numerical evaluation of the ionization integrals, is the breakdown voltage.

(a) Paper (original)



(b) This reproduction

Figure 1: **Fig. 3 — equipotential contours** ($t_s = 0.1\,\mu\text{m}$, $t_{ox} = 2\,\mu\text{m}$, $L = 70\,\mu\text{m}$), at the breakdown voltage. The model plots the closed-form $\phi(x, y)$ directly — no numerical Poisson solve is needed, since the paper's whole design is a closed-form field. The design width $W = 49.4\,\mu\text{m}$ is found by inverting $\text{BV}(W) = 750\,\text{V}$ to match the paper's stated result.

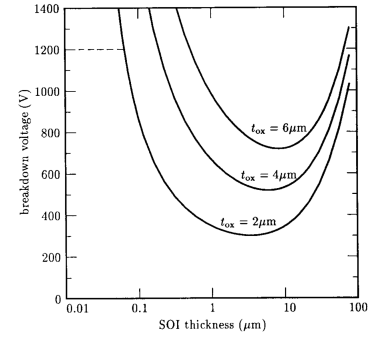
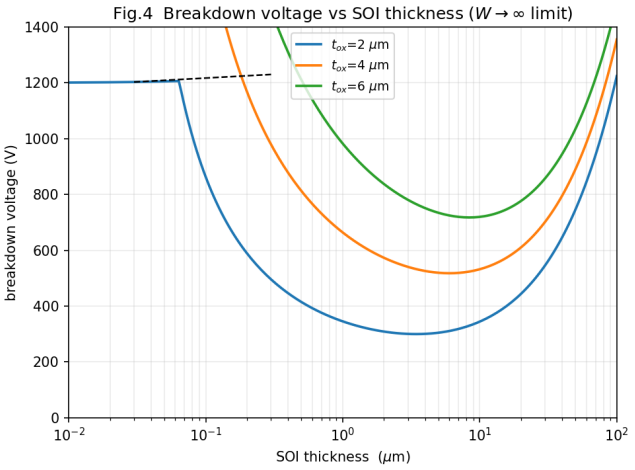


FIG. 4: Theoretical breakdown voltage versus SOI thickness, for three values of buried oxide thickness t_{ox}. This graph is valid in the limit where the lateral depletion width is large. The dashed line corresponds to dielectric breakdown of the 2 μm-thick buried oxide.

(a) Paper (original)



(b) This reproduction

Figure 2: **Fig. 4 — breakdown voltage vs. SOI thickness** ($W \to \infty$ limit), for $t_{ox} = 2/4/6\,\mu\text{m}$. The characteristic "U" shape and the oxide-breakdown cap (1210 V modeled vs. 1200 V paper) reproduce closely. The absolute BV level runs ∼15% higher than the paper's text ("700 V may be achieved"); since Fig. 5's plateau and the oxide cap both match closely, this is most likely the paper's rounded reference to its >700 V experimental headline rather than a precise Fig. 4 graph reading.

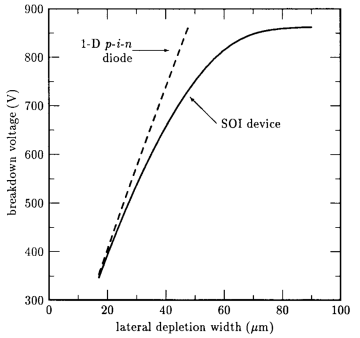
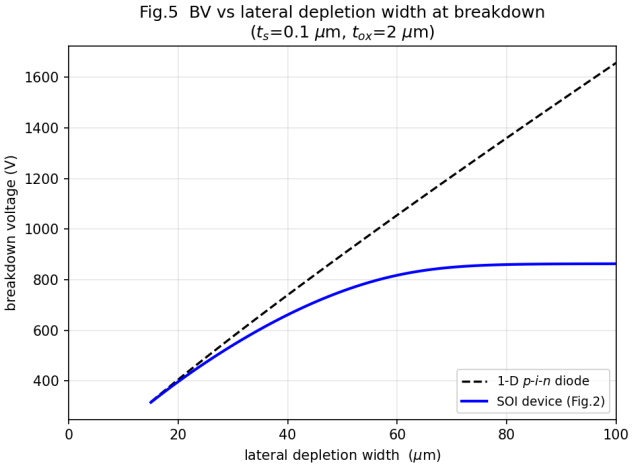


FIG. 5: Theoretical breakdown voltage versus lateral depletion width at breakdown. The solid curve represents the SOI device of Fig. 2, where $t_s = 0.1$ μm and $t_{ox} = 2$ μm. The dashed curve represents the one-dimensional p-i-n diode.

FIG. 6
diode.
5 μA
referri



(a) Paper (original)

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Figure 3: **Fig. 5 — breakdown voltage vs. lateral depletion width** W ($t_s = 0.1\,\mu\text{m}$, $t_{ox} = 2\,\mu\text{m}$), vs. the 1-D p-i-n reference. Excellent match: both the SOI-device saturation (∼863–870 V) and the linear p-i-n reference track the paper closely over the full range.

3 Extension — 4H-SiC and GaN films

Nothing in Sec. 1 is silicon-specific except the single-carrier Chynoweth ionization fit and the film's relative permittivity K_s (the buried oxide stays SiO_2, $K_{ox} = 3.9$). Swapping in literature Chynoweth fits for 4H-SiC and GaN re-evaluates the *same* closed-form model (`model.py`'s `Material/MATERIALS` registry) for a wide-bandgap-on-insulator version of the device.

Material	K_s	a (cm^{-1}), b ($\mathrm{V\,cm}^{-1}$)	Source
Si	11.7	7.03×10^5, 1.468×10^6	Merchant et al. 1991, eq. 10
4H-SiC	9.7	3.25×10^6, 1.71×10^7	Raghunathan & Baliga, APL **72**, 3503 (1998), holes
GaN	8.9	6.4×10^5, 1.454×10^7	Cao et al., APL **112**, 262103 (2018), holes

3.1 Result: the oxide, not the film, limits at this design point

At the paper's own design point ($t_s = 0.1\,\mu\mathrm{m}$, $t_{ox} = 2\,\mu\mathrm{m}$, $W \to \infty$), 4H-SiC's and GaN's much larger b (higher ionization threshold field) implies enormous semiconductor-limited breakdown voltages if the film alone is considered:

Material	Semiconductor-limit BV	Oxide cap	Actual (capped) BV	Oxide/film crossover t_s
Si	863 V	1210 V	863 V	0.064 µm
4H-SiC	3990 V	1212 V	1212 V	32.8 µm
GaN	6947 V	1213 V	1213 V	28.8 µm

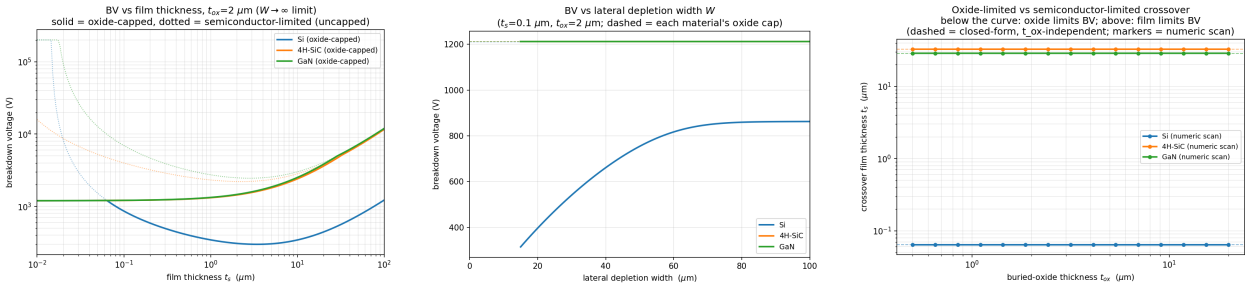
But the buried SiO_2 breaks down first for both wide-bandgap materials: its dielectric strength ($600\,\mathrm{V\,\mu m}^{-1}$, unchanged) caps all three materials at essentially the same ∼1210–1213 V, since the oxide cap is set almost entirely by t_{ox} once $t_s \ll t_{ox}$. **In this architecture, at this film thickness, 4H-SiC and GaN gain nothing over silicon.**

3.2 A closed-form, t_{ox}-independent crossover

Both BV (the $W \to \infty$ semiconductor limit) and the oxide breakdown voltage V_{ox} are proportional to the same $t = t_s/2 + (K_s/K_{ox})t_{ox}$: $\mathrm{BV} = bt/\gamma^*(t_s)$, where $\gamma^*(t_s)$ is the root of eq. (7)'s $I_2' = 1$ condition, and $V_{ox} = E_{ox,\mathrm{crit}}\, t/(K_s/K_{ox})$. Setting $\mathrm{BV} = V_{ox}$, the common factor t *cancels*, leaving a condition on t_s alone:

$$\gamma_{\mathrm{cross}} = \frac{b\,(K_s/K_{ox})}{E_{ox,\mathrm{crit}}}, \qquad a\,t_s\,\gamma_{\mathrm{cross}}\left[\mathrm{Ei}(-\gamma_{\mathrm{cross}}) + \frac{e^{-\gamma_{\mathrm{cross}}}}{\gamma_{\mathrm{cross}}}\right] = 1. \tag{9}$$

The oxide/film crossover thickness is exactly independent of t_{ox} — a closed-form fact of the model (`model.crossover_film_thickness`), not a numerical coincidence, confirmed by brute-force root-finding across t_{ox} in Fig. 4. Silicon's crossover (0.064 µm) sits right at the edge of the paper's own thin-film regime, which is why Si's Fig. 4 "U" is genuinely film-limited over most of its interesting range. 4H-SiC's and GaN's crossover thicknesses (∼29–33 µm) are two orders of magnitude larger: **the whole point of this paper — pushing the film to sub-micron thickness to raise BV — does not transfer to wide-bandgap films with a plain 2 µm SiO_2 BOX**, because at any sub-micron thickness they are deep in the oxide-limited regime. Exploiting 4H-SiC's or GaN's intrinsic advantage in this architecture would require either a much thicker buried oxide or a higher-dielectric-strength insulator, not a thinner film.



(a) BV vs. film thickness, $t_{ox} = 2\,\mu\text{m}$ (b) BV vs. lateral depletion width W (c) Oxide/film crossover t_s

Figure 4: **4H-SiC / GaN extension** (model-only; no corresponding paper figure). (a) Solid = oxide-capped (actual), dotted = semiconductor-limited (uncapped). (b) $t_s = 0.1\,\mu\text{m}$, $t_{ox} = 2\,\mu\text{m}$; dashed = each material's oxide cap. (c) Crossover film thickness t_s is exactly independent of t_{ox} for all three materials (closed form matches brute-force numerical scan).

4 The plateau problem — why Si's BV(t_s) minimum is broad

Fig. 4's "U" curve has a minimum in t_s; visually, silicon's is a broad, flat plateau while 4H-SiC's and GaN's are narrow, sharp "V"s. This follows from an exact factorization of the model already used above:

$$V_{BR}(t_s, t_{ox}) = E_c(t_s) \cdot t_{eff}(t_s, t_{ox}), \qquad E_c(t_s) := \frac{b}{\gamma^*(t_s)}, \tag{10}$$

where $\gamma^*(t_s)$ is the root of $a\,t_s\,\gamma\big[\mathrm{Ei}(-\gamma) + e^{-\gamma}/\gamma\big] = 1$ — an equation in t_s and the material's ionization prefactor a *alone* (`model.g_star`, `model.critical_field`); neither t_{ox} nor K_s enters $E_c(t_s)$ at all (verified numerically to solver tolerance). Taking logarithmic derivatives, $dV_{BR}/dt_s = 0$ where

$$\underbrace{-\frac{d\ln E_c}{d\ln t_s}}_{\text{material \& } t_s \text{ only}} \quad = \quad \underbrace{\frac{t_s/2}{t_s/2 + r\,t_{ox}}}_{\text{material's } r,\ t_s,\ t_{ox}}, \qquad r = \frac{K_s}{K_{ox}}. \tag{11}$$

This crossing was checked against a direct numerical minimization of $V_{BR}(t_s)$ for all 9 (material, t_{ox}) combinations tested: it locates the minimum to within 0.01 µm every time.

Why Si's plateau is broader. The LHS of (11) depends on material only through a (not b, not K_s). Si's Chynoweth b is ∼10× smaller than 4H-SiC's/GaN's, so its critical field's log-slope varies far more gently with t_s; a gently-varying LHS stays nearly matched to the slowly-varying, bounded RHS over a *wide* t_s range (broad plateau), while a steeply-varying LHS (4H-SiC, GaN) crosses sharply at one point (narrow "V").

Quantified. Defining the plateau width as the t_s-interval around the numerically-found minimum over which $|dV_{BR}/dt_s|/V_{BR,\min}$ stays below 1%/µm:

t_{ox}	Si	4H-SiC	GaN
2 µm	1.19 µm	0.58 µm	0.73 µm
4 µm	4.47 µm	2.27 µm	2.78 µm
6 µm	13.03 µm	5.24 µm	6.34 µm

Si's plateau is consistently ∼2× wider than 4H-SiC's or GaN's at the same t_{ox}; permittivity (entering only through the RHS) is a secondary effect, the dominant cause being Si's much weaker field-sensitivity of impact ionization (b), not permittivity.

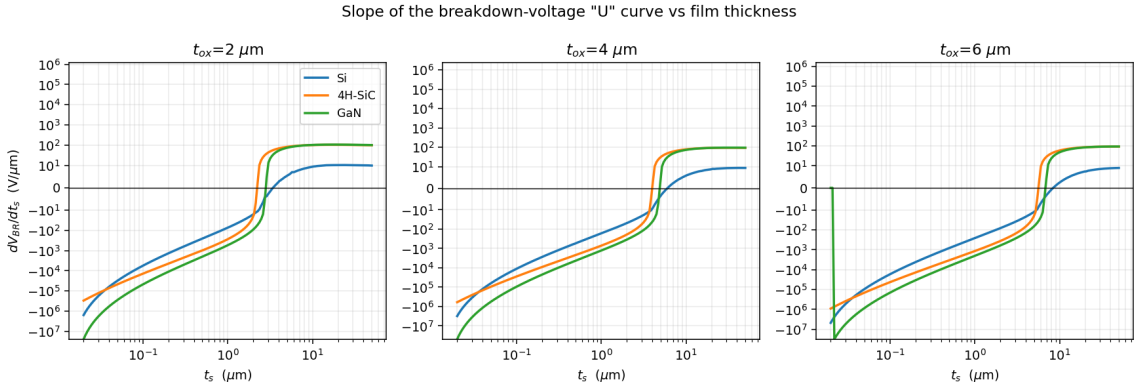


Figure 5: **Raw derivative dV_{BR}/dt_s vs.** t_s, Si/4H-SiC/GaN, one panel per $t_{ox} \in \{2, 4, 6\}$ µm. The wider the flat region around the zero-crossing, the broader the plateau.

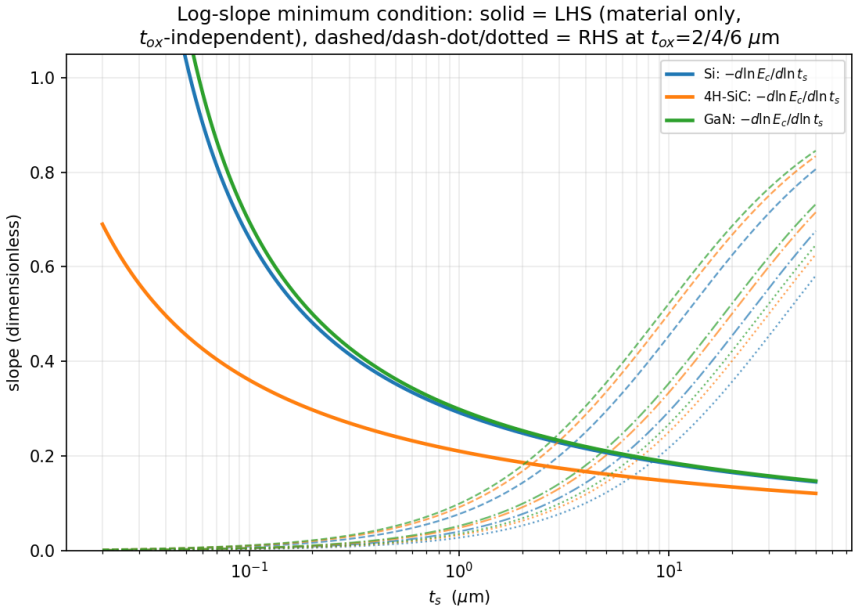


Figure 6: **Log-slope minimum condition** (11): solid = LHS (material-only, t_{ox}-independent); dashed/dash-dot/dotted = RHS at $t_{ox} = 2/4/6\,\mu$m. Crossings match $V_{BR}(t_s)$'s numerical minima to within 0.01 µm.

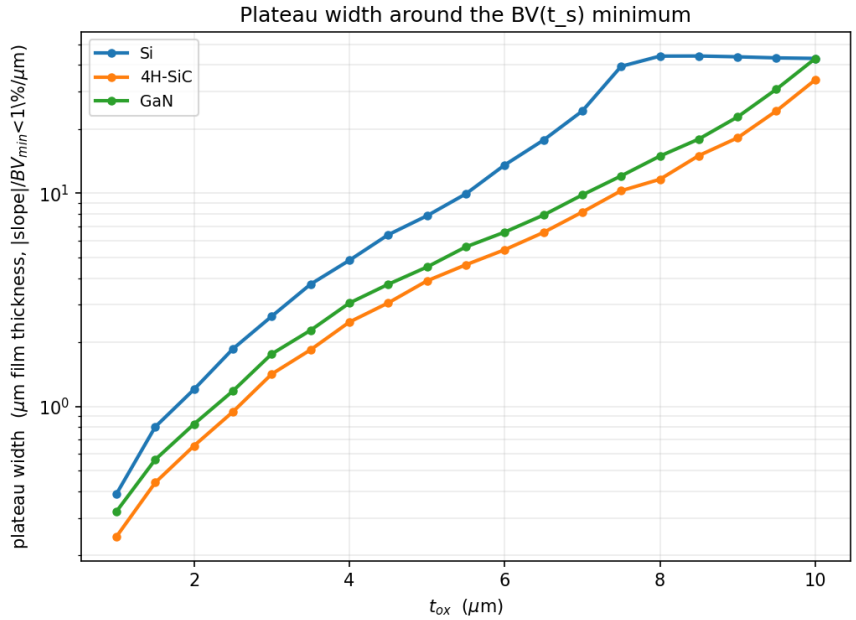


Figure 7: **Plateau width vs.** t_{ox}, per material — Si is consistently ∼2× wider than 4H-SiC or GaN at the same t_{ox}.

5 Notes and limitations

Fig. 6 (experimental I-V trace) — out of scope. Fig. 6 is a photographed oscilloscope trace of a fabricated device; it is experimental data, not a simulation target (`figs_paper/fig6_paper_crop.png` holds the original).

Known quantitative gap. The absolute breakdown-voltage level in Fig. 4 runs ∼15% higher than the paper's text; however Fig. 5's $W \to 100\,\mu\text{m}$ plateau (∼865 V) and the oxide-breakdown cap (1210 vs. 1200 V) both match closely, so the paper's "700 V" text is most likely a rounded reference to the >700 V experimental headline rather than a precise Fig. 4 graph reading.

Assumptions. Ionization coefficients are the paper's own stated fit (eq. 10), not the Van Overstraeten–de Man values used in later papers. Only the two paper-identified dominant ionization paths are evaluated (eq. 13's product formally has infinitely many paths). Fig. 3's device is assumed to follow the derived linear $N(x)$ out to a single design width W; beyond W the SOI is treated as quasi-neutral at the applied bias. The 4H-SiC and GaN fits use hole ionization coefficients for comparability across materials — see `model.py`'s docstring for the full reasoning.

Code: `model.py` — core closed-form fields, ionization integrals, and the `Material` registry; `fig3_potential.py`, `fig4_bv_vs_ts.py`, `fig5_bv_vs_W.py` — one script per reproduced paper figure; `fig_materials_comparison.py` — the 4H-SiC/GaN extension and crossover analysis; `fig_plateau_analysis.py` — the dV_{BR}/dt_s, log-slope, and plateau-width figures of Sec. 4.